

The discrete maximal operator blocks direct extrapolation below one

Candidate full-counterexample packet for arXiv:2304.14250

11 August 2026

Status: candidate full counterexample to the direct interpretation, likely valid, subject to human review. The discrete Hardy–Littlewood maximal operator satisfies the source’s weighted hypothesis at every fixed $p_0 > 1$, but is not bounded on unweighted $\ell^p(\mathbb{Z}_+)$ for any $0 < p \leq 1$. Therefore no below-one extension of the source theorem that includes the constant weight can be true under the same hypothesis.

1 The signal and its necessary interpretation

Saker, Saied, and Agarwal prove discrete Rubio de Francia extrapolation for $1 < p < \infty$ and say in the abstract and Introduction that the case $0 < p < 1$ is open [1]. Their discrete Muckenhoupt class $\mathcal{A}_p$ is defined only for $p > 1$, so a literal conclusion with “ $w \in \mathcal{A}_p$ ” is not defined below one. Nevertheless, every direct extension of their operator theorem has one unavoidable consequence: it must give the unweighted ℓ^p estimate, since the constant weight is the baseline weight at every classical exponent.

That necessary consequence is false. Thus the result below answers the only direct interpretation of the printed signal. It also diagnoses why a valid below-one theorem must change either the initial hypothesis, the target weight class, the sequence family, or the target norm.

Idea of the proof

Use the operator that generates the weight classes in the first place. The discrete Hardy–Littlewood maximal operator obeys the required weighted strong-type estimate for every $p_0 > 1$ and every $\mathcal{A}_{p_0}$ weight. But on the point mass e_1 , the interval $\{1, \dots, n\}$ gives $Me_1(n) \geq 1/n$. This tail is outside ℓ^p exactly when $p \leq 1$.

2 The counterexample

For a sequence f on $\mathbb{Z}_+ = \{1, 2, \dots\}$, let

$$Mf(n) = \sup_{\substack{I \subset \mathbb{Z}_+ \\ \text{finite interval} \\ n \in I}} \frac{1}{|I|} \sum_{k \in I} |f(k)|$$

be the discrete uncentered Hardy–Littlewood maximal operator. The same argument works for any one-sided variant which takes the initial intervals $\{1, \dots, n\}$ among its averages.

Theorem 1. Fix $p_0 > 1$. The operator M satisfies the source hypothesis

$$\sum_{n=1}^{\infty} (Mf(n))^{p_0} w(n) \leq C(p_0, [w]_{\mathcal{A}_{p_0}}) \sum_{n=1}^{\infty} |f(n)|^{p_0} w(n) \quad (1)$$

for every discrete $\mathcal{A}_{p_0}$ weight w . However, for every $0 < p \leq 1$, there is no constant C_p such that

$$\sum_{n=1}^{\infty} (Mf(n))^p \leq C_p \sum_{n=1}^{\infty} |f(n)|^p$$

for all finitely supported f .

Proof. The weighted strong-type estimate (1) is the defining maximal theorem for discrete Muckenhoupt weights. It is recalled explicitly in the source and proved in the discrete setting by Saker and Mahmoud [2]. Replacing its dependence on the characteristic by a monotone envelope, if necessary, gives exactly the positive increasing control function required by the source extrapolation hypothesis.

Let e_1 be the point mass at 1. For every $n \geq 1$, the interval $I_n = \{1, \dots, n\}$ contains n , and hence

$$Me_1(n) \geq \frac{1}{n} \sum_{k=1}^n e_1(k) = \frac{1}{n}.$$

Consequently, for $0 < p \leq 1$,

$$\sum_{n=1}^{\infty} (Me_1(n))^p \geq \sum_{n=1}^{\infty} \frac{1}{n^p} = \infty, \quad \sum_{n=1}^{\infty} |e_1(n)|^p = 1.$$

Thus M is not bounded on unweighted $\ell^p(\mathbb{Z}_+)$ at or below one, although it satisfies the full weighted p_0 hypothesis. $\square$

Corollary 1 (family-of-pairs form). *Let*

$$\Omega = \{(Mf, f) : f \geq 0 \text{ and finitely supported on } \mathbb{Z}_+\}.$$

At exponent $p_0 > 1$, the source's weighted inequality holds for every pair in Ω and every $w \in \mathcal{A}_{p_0}$. For every $0 < p \leq 1$, its unweighted conclusion fails for the pair (Me_1, e_1) .

This removes a possible operator-linearity loophole: the source emphasizes that its theorem is about arbitrary families of pairs and does not require linearity or sublinearity. The counterexample already works in that generality.

3 Scope, stronger variants, and novelty

The counterexample does not contradict the source's final $\mathcal{A}_{\infty}$ corollary. That corollary assumes the initial p_0 inequality for *every* $\mathcal{A}_{\infty}$ weight, a strictly stronger premise than the main $\mathcal{A}_{p_0}$ hypothesis. Nor does it contradict extrapolation for decreasing or quasi-nonincreasing sequences with Ariño–Muckenhoupt-type weight classes. A 2026 paper proves an upward extrapolation theorem for a restricted quasi-nonincreasing family and a different $\mathcal{QB}_{\beta,p}$ class [3]; it does not supply the direct all-sequence $\mathcal{A}_p$ extension refuted here.

Accordingly, a meaningful below-one replacement must be reformulated. The source's own $\mathcal{A}_{\infty}$ corollary is one option; restricted monotonicity, B_p -type classes, weak-type targets, or Hardy-space

substitutes are other possible directions. The present theorem rules out retaining only the main $\mathcal{A}_{p_0}$ premise while demanding an ℓ^p conclusion that includes the constant weight.

On 11 August 2026, the run registry and all cheap indexes had no packet for arXiv:2304.14250. Exact local-corpus and bounded primary-source searches found the source and the later restricted-sequence paper, but no resolution of this exact signal. This is a bounded novelty check, not an exhaustive priority claim. The obstruction itself is elementary and should be treated as a clarification/counterexample rather than a claim of deep novelty.

A Source-page evidence

Figure 1 reproduces the source abstract, including the below-one open signal.

References

- [1] S. H. Saker, A. I. Saied, and R. P. Agarwal, *Discrete Rubio de Francia extrapolation theorem via factorization of weights and iterated algorithms*, arXiv:2304.14250 [math.FA], 2023.
- [2] S. H. Saker and R. R. Mahmoud, *Boundedness of both discrete Hardy and Hardy–Littlewood maximal operators via Muckenhoupt weights*, Rocky Mountain J. Math. 51 (2021), 733–746.
- [3] M. Singh, A. Gogatishvili, R. Panchal, and A. P. Singh, *Rubio de Francia extrapolation theorem for quasi non-increasing sequences*, arXiv:2603.06109 [math.FA], 2026.

DISCRETE RUBIO DE FRANCIA EXTRAPOLATION THEOREM VIA FACTORIZATION OF WEIGHTS AND ITERATED ALGORITHMS

S. H. SAKER¹, A. I. SAIED² AND R. P. AGARWAL³

ABSTRACT. In this paper, we prove a discrete Rubio de Francia extrapolation theorem via factorization of discrete Muckenhoupt weights and discrete iterated Rubio de Francia algorithm and its duality. Precisely, we will prove that if for some $p_0 > 1$ and $w \in \mathcal{A}_{p_0}$ the $\ell^{p_0}(w)$ inequality $\|\mathcal{T}f\|_{\ell^{p_0}(w)} \leq C \|f\|_{\ell^{p_0}(w)}$ holds for pairs of the form $(\mathcal{T}f, f)$, then for all p and $w \in \mathcal{A}_p$ the $\ell^p(w)$ inequality $\|\mathcal{T}f\|_{\ell^p(w)} \leq C \|f\|_{\ell^p(w)}$ also holds for such pairs, where $\mathcal{T}$ is a given operator. The new approach, among other things, allows us to obtain explicit estimates of the constants and do not assume that the operator $\mathcal{T}$ is linear or even sublinear. The theorem will be proved for a fixed exponent $1 < p < \infty$ and the case when $0 < p < 1$ is an open problem.

2010 Mathematics Subject Classification: 44A55, 26D15, 40A05, 39A70, 46B45.

Key words and phrases: Extrapolation theorem, Muckenhoupt classes, Bounded operator and Rubio de Francia algorithm.

1. INTRODUCTION

The study of boundedness of operators is one of the main subjects in modern harmonic analysis. The ideas related to the boundedness of operators and the weighted inequalities appeared with the birth of singular integrals but a better understanding of the subject was only in the 1970s. In 1972 Muckenhoupt [18] introduced the A_p -weights (now often referred to as Muckenhoupt weights): for a given exponent $p > 1$ a weight w is said to belong to A_p if there exists a real constant $C > 1$ such that,

$$(1.1) \quad \left(\frac{1}{|I|} \int_I w(x) dx \right) \left(\frac{1}{|I|} \int_I w^{-\frac{1}{p-1}}(x) dx \right)^{p-1} \leq C,$$

holds for every interval $I \subseteq \mathbb{R}$. The infimum of all such C is denoted by $[w]_{A_p}$ and is called the A_p -norm. The full characterization of the weights w , for which the Hardy-Littlewood maximal operator

$$(1.2) \quad \mathcal{M}f(x) := \sup_{x \in I} \frac{1}{|I|} \int_I f(y) dy,$$

is bounded on $L^p_w(\mathbb{R}^+)$ by means of the so-called A_p -condition was achieved by Muckenhoupt and the weighted norm inequality

$$(1.3) \quad \int_0^\infty w(t) (\mathcal{M}f(t))^p dt \leq C \int_0^\infty w(t) f^p(t) dt.$$

has been proved for a nonnegative function $f \in L^p_w(\mathbb{R}^+)$ and the constant C depends of the norm $[w]_{A_p}$ of the weight w . The weight w is in the Muckenhoupt class A_1 if there exists a constant $C > 1$ such that

$$(1.4) \quad \mathcal{M}w(x) \leq Cw(x), \quad \text{for all } x \in \mathbb{R}^+,$$

where the operator $f \rightarrow \mathcal{M}f$ is the Hardy-Littlewood maximal operator. The infimum of all such C is denoted by $[w]_{A_1}$ and is called the A_1 -norm.

Figure 1: The first printed source page: the abstract states that the case $0 < p < 1$ is an open problem.